



# ABDELHAK BOURDIM

Senior Embedded Software Engineer  
C/C++ • RTOS • Embedded Linux

## CONTACT

✉ abdelhak.bourdim@gmail.com

☎ +33 6 19 51 51 73

🚩 Paris area, France

🌐 linkedin.com/in/abdelhak-bourdim-96416122

⚙️ workshop-diy.org

## LANGUAGES

French ●●●●● *Fluent*  
English ●●●●● *Professional*  
Arabic ●●●●● *Fluent*

## STANDARDS & METHODS

DO-178 MISRA C IEC 62304  
ISO 15118 (V2G)  
ISO 14443 (Smart Card)  
ISO 15693 (RFID) OCPP V-Model  
Agile / Scrum

## SKILLS

LANGUAGES  
C, C++, Python, Bash, Assembly, Perl, Awk

OS / RTOS  
Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel

MICROCONTROLLERS  
ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto, Arduino

PROTOCOLS / BUS  
CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP

STACKS  
TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer

SECURITY / PAYMENT  
Smart Card, RFID Mifare, DES, RC4, USB CCID

TOOLS  
Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx,

## PROFILE

Embedded software specialist — firmware, middleware and application layers, bare metal, RTOS or Embedded Linux — for critical and industrial systems, with **over 20 years of experience** in driver development, communication stacks (CAN, BLE, USB, TCP/IP), RTOS and Embedded Linux on ARM Cortex, STM32, Microchip and NXP architectures.

Delivered certified embedded solutions for **aerospace** (Safran, Thales — DO-178), **medical** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT cybersecurity** (Enedis — Linky smart meters), **payment systems** (Cartadis) and **V2G smart charging** (E2-CAD — ISO 15118). MISRA C compliant development. Founder of **Workshop-DIY**, an educational robotics non-profit having led 18+ workshops for youth (Arduino, ESP32, AI, 3D printing).

## PROFESSIONAL EXPERIENCE

### E2-CAD — Eragny

May 2025 – Present

#### EV Charging Station — Iotecha IP

Product takeover and test bench setup

- ▶ Reverse engineering and source code analysis (static and dynamic architecture)
- ▶ Embedded code review on 3 boards (Microchip PIC, STM32), setup of debug and programming tools (ST-Link, nsprog)
- ▶ Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, frame extraction with Saleae)
- ▶ Python extension development and real-time HTML web report generation
- ▶ Test bench design and setup: Chargebyte tools integration, EV simulator
- ▶ Charging station debug and commissioning (PLC, HomePlug Green PHY, ISO 15118, OCPP)

Env: Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, Chargebyte, RS232, I2C, CAN, TCP/IP, Saleae, HackRF One, SSH, SCP, Wireshark, Python, Bash, HTML

### Safran — Créteil

Mar. 2024 – May 2025

#### SSPC Project — Solid State Power Controller

Low-level driver updates

- ▶ Development and update of low-level C drivers for the SSPC system
- ▶ Implementation of logging tools for diagnostics and exchange traceability
- ▶ Writing and execution of on-target validation tests with Lauterbach TRACE32
- ▶ Bash/Awk scripting for test process automation

Env: C, Lauterbach TRACE32, Bash, Awk

### Asterios Technologies — Massy

Aug. 2023 – Feb. 2024

#### T1042SOM Project — Software Tests

Software test procedure writing

- ▶ C development of validation tests for the SOM module (T1042 processor)
- ▶ Processor errata analysis and software workaround identification
- ▶ Test documentation writing compliant with project requirements

Env: C, Bash, Python, Lauterbach TRACE32, Polarion

### Thales — Gennevilliers

May 2022 – Jul. 2023

#### LPM + N-MMR Projects — Defense Radio

Mission preparation software & New Multi Mode Receiver

- ▶ Application layer development under Embedded Linux in C/C++ for 2 projects
- ▶ Configuration management and automation scripts (Bash, Curl, Tshark)
- ▶ Development and execution of HLT (High Level Tests)
- ▶ Bug fixing and network protocol analysis (Websockets)

Env: Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

### Valeo — Créteil

Mar. – Apr. 2022

#### Test Audit — Valeo Egypt

Unit test, integration test and tools audit

- ▶ Audit of test processes and tools (NI Mastre, Vector CANoe)
- ▶ Automation and execution time improvement proposals
- ▶ Client/system requirements review, audit report writing

Env: NI Mastre, Vector CANoe

### Arkamys — Levallois-Perret

Jul. 2021 – Feb. 2022

#### GStreamer Framework — Embedded Audio

Audio plugins, unit & integration tests, End-to-End tests

- ▶ GStreamer plugin and CAN driver (ELM327) development, diagnostic tools
- ▶ Unit & integration testing (Valgrind), build system Buildroot

Env: Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

INDUSTRY SECTORS

- ✈ Aerospace — Safran, Thales, Zodiac
- 🏥 Medical — GE Healthcare, Sorin
- ⚙ Automotive — Valeo, Arkamys
- ⚡ Energy — Enedis (Linky)
- ⚡ EV Charging / V2G — E2-CAD
- 📶 IoT / M2M — Vertical M2M
- 💳 Payment — Cartadis, Photomaton, BCM

KEY STRENGTHS

- ✓ Embedded Linux & RTOS
- ✓ Embedded cybersecurity (pentest, audit)
- ✓ Embedded software development
- ✓ Communication protocols
- ✓ Testing & validation
- ✓ Diagnostics & debugging

NON-PROFIT PROJECT

Workshop-DIY

FOUNDER — CHELLES (77)

Robotics & STEM workshops for youth (ages 6-12+)  
ESP32 robot kit design (WiFi, BLE, servos, sensors)  
18+ workshops: Arduino, micro:bit, AI, 3D printing, laser  
workshop-diy.org

EDUCATION & TECH WATCH

- Electronic & Computer Systems Design Engineer  
AFPA, Champs-sur-Marne
- Biomedical Instrumentation (Postgraduate)  
Tenon Hospital, Paris
- Electronics Engineer, Communication  
Boumerdès, Algeria

TECHNOLOGY WATCH

- ▶ Embedded cybersecurity & IoT pentest
- ▶ V2G / ISO 15118 / Smart Charging
- ▶ SDR & RF protocol analysis
- ▶ Embedded AI & Edge Computing

PROFESSIONAL EXPERIENCE (CONTINUED)

Thales — Vélizy-Villacoublay

Nov. 2020 — Jun. 2021

ATO — Autonomous Train Operation

IPC communication & socket programming

- ▶ Application migration from 32-bit to 64-bit on Embedded Linux platform
- ▶ IPC communication protocol definition and implementation
- ▶ Network diagnostic tools development and integration testing
- ▶ Frame analysis with Wireshark, Tshark, Pscapy and Pyshark

Env: Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Pscapy, Python, Jira

PK Vitality — Paris

Jun. — Sep. 2020

K'Watch CGM — Continuous Glucose Monitor

BLE drivers, FreeRTOS, STM32 TouchGFX

- ▶ HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- ▶ STM32 TouchGFX touchscreen, FreeRTOS, low power mode
- ▶ In vitro testing and Doxygen documentation (IEC 62304 compliant)

Env: STM32L4S9, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen

Zodiac Aerospace (Safran) — Montreuil

Feb. 2017 — Mar. 2020

GENOME / CORAC / SSPC A350 AC9

Low-level drivers, diagnostics, application layers

- ▶ Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- ▶ uAFDX to CAN gateway implementation and diagnostic functions
- ▶ SSPC A350 application layer development and optimization
- ▶ Validation testing and aerospace standards-compliant documentation

Env: Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C

Sorin — Clamart

Nov. 2014 — Apr. 2015

Platinum — BLE Module for Pacemaker

Bluetooth integration feasibility study for cardiac implant

- ▶ Bluetooth Smart module validation, Serial Port Service Profile development
- ▶ Smartphone communication protocol development and Android application
- ▶ Hardware interface design and sample preparation for implantation

Env: Cortex M0, Keil, C, Android (Eclipse)

General Electric Healthcare — Buc

2012 — 2014

ZYX — X-Ray Generator

CANOpen integration on RCB and CORTYX boards

- ▶ CANOpen stack integration (slave/master), CAN driver and service development
- ▶ CMX-RTX kernel porting, file management, TCL test scripts
- ▶ USA team technical support and on-site visits in the United States

Env: TI TMS320, XE167, ARM Cortex A8, VxWorks, C, C++

PREVIOUS EXPERIENCE

Enedis — Nanterre

May 2015 — Sep. 2016

Linky Smart Meters — Smart Grids

Cybersecurity and Linky concentrator audit

- ▶ Linky concentrator security diagnostics and audit
- ▶ Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- ▶ Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)
- ▶ Log analysis and batch firmware update tool development
- ▶ Code audit and review with manufacturers (Sonar, PC Lint)

Env: Kali Linux, Embedded Linux, Nmap, Wireshark, Tshark, Scapy, SDR, Saleae, Python, Bash, Arduino, Raspberry Pi

Photomaton / BCM / Vertical M2M / PGS — Paris

2010 — 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

Cartadis — Fontenay-sous-Bois

Oct. 2001 — Dec. 2009

8 years — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)